

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

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DAT313

Cost-optimize SQL Server workloads on Amazon RDS and Amazon RDS Custom

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Audience poll



Agenda

- 01 Review of Amazon RDS and RDS Custom for SQL Server
- 02 General cost optimization strategies for RDS SQL Server and RDS Custom for SQL Server
- 03 Cost-optimize RDS and RDS Custom instances, storage, and Auto Scaling
- 04 Discussion

1. Review of Amazon RDS and RDS Custom for SQL Server

SQL Server deployment options

SQL Server on Amazon EC2

Self-managed experience

Full control

All DB engine features

Self-managed provisioning, monitoring, backup, restore, point-in-time recovery

Self-managed patching

Self-managed high availability

Up to 5,000 databases

Allows third-party applications on DB host

BYOL, license included

All

All

RDS Custom for SQL Server

Managed experience (shared responsibility)

Full control

Access all SQL Server configurations

Automated provisioning, monitoring, backup, restore, point-in-time recovery

Automated patching*

Automated high availability

Up to 5,000 databases

Allows third-party applications on DB host

BYOM, license included

Web, Standard, Enterprise, Developer

2019

RDS for SQL Server

Managed experience

No sysadmin/OS access

Optimized architecture

Automated provisioning, monitoring, backup, restore, point-in-time recovery

Automated patching

High availability & Multi-AZ

Up to 100 databases

No third-party applications on DB host

License included

Express, Web, Standard, Enterprise

2014, 2016, 2017, 2019, 2022

Self-managed

AWS managed



Amazon RDS Custom use cases



Granular control

Install custom drivers, enable features or applications that require elevated privileges

Example: Extended stored procedures, CLR, resource governor, linked server (various DB engines)



Lift and shift business applications

Third-party or packaged applications with minimal changes

Example: Microsoft SharePoint, Microsoft Dynamics



Disaster recovery

Set up DR from a self-managed environment

Example: SQL Server Always On availability groups, replication

2. General cost optimization strategies

Understanding the RDS cost model

Traditional RDBMS cost model

- Provision for peak usage, pay for peak usage (regardless of actual use)
- Likely provision an additional buffer in addition to the peak
- Example: Provisioned 100GB in anticipation of database growth, but current utilization may be only 10GB

RDS cost model

- RDS provides flexibility to provision resources as needed
- Only pay for what you use
- **Scale storage** automatically
- **Scale CPU** by schedule

Cost optimization strategies

Cost monitoring

Compute optimization

Storage optimization

SQL Server Edition optimization

Data transfer optimization

Database consolidation

Consolidation – Example workload

	E	F	G	I	P	BL	BM	BN	BO
1									
2				Total Monthly Compute Costs			\$ 103,649.05		\$ 94,410.90
3									
4				Total Monthly Savings (USD)			\$ 9,238.15		
5									
6				Total Monthly Savings (%)			8.91%		
7									
8	Environment	DbInstanceId.options.label	Engine	AverageCPUUtil	details.dbInstanceClass	details.MaximumCPU	Estimated Current	RightSizedInstanceC	Estimated Right-Size
12	Development	db-mssql-biztalk-mule-use1-dev	sqlserver-ee 15.00.4043.16.v1	2.27%	db.r5.xlarge	15.86%	\$ 1,825.73	db.r5.xlarge	\$ 1,825.73
48	Development	db-mssql-erpm-use1-dev	sqlserver-ee 15.00.4043.16.v1	2.55%	db.r5.xlarge	17.43%	\$ 1,825.73	db.r5.xlarge	\$ 1,825.73
49	Development	db-mssql-financertpg-use1-dev	sqlserver-ee 12.00.6329.1.v1	1.98%	db.r5.xlarge	33.56%	\$ 1,825.73	db.r5.xlarge	\$ 1,825.73
50	Development	db-mssql-shared-euw1-009-dev	sqlserver-ee 15.00.4043.16.v1	3.38%	db.r5.xlarge	33.74%	\$ 1,825.73	db.r5.xlarge	\$ 1,825.73
51	Development	db-mssql-shared-use1-003-dev	sqlserver-ee 15.00.4043.16.v1	2.81%	db.r5.xlarge	7.80%	\$ 1,825.73	db.r5.xlarge	\$ 1,825.73
52	Development	db-mssql-shared-use1-007-dev	sqlserver-ee 15.00.4043.16.v1	3.63%	db.r5.xlarge	83.08%	\$ 1,825.73	db.r5.xlarge	\$ 1,825.73
53	Development	db-mssql-shared-use1-011-dev	sqlserver-ee 13.00.6300.2.v1	3.41%	db.r5.xlarge	50.57%	\$ 1,825.73	db.r5.xlarge	\$ 1,825.73
54	Development	db-mssql-shared-use1-013-dev	sqlserver-ee 12.00.6329.1.v1	2.53%	db.r5.xlarge	23.77%	\$ 1,825.73	db.r5.xlarge	\$ 1,825.73
55	Development	db-mssql-shared-use1-020-dev	sqlserver-ee 15.00.4043.16.v1	2.69%	db.r5.xlarge	35.83%	\$ 1,825.73	db.r5.xlarge	\$ 1,825.73
56	Development	db-mssql-shared-use1-023-dev	sqlserver-ee 15.00.4043.16.v1	2.35%	db.r5.xlarge	17.44%	\$ 1,825.73	db.r5.xlarge	\$ 1,825.73
57	Development	db-mssql-shared-use1-029-dev	sqlserver-ee 15.00.4043.16.v1	2.42%	db.r5.xlarge	19.97%	\$ 1,825.73	db.r5.xlarge	\$ 1,825.73
124	Development	db-mssql-shared-usw2-001-dev	sqlserver-ee 15.00.4043.16.v1	2.45%	db.r5.xlarge	18.04%	\$ 1,825.73	db.r5.xlarge	\$ 1,825.73
130	Development	db-mssql-shared-usw2-019-dev	sqlserver-ee 12.00.6329.1.v1	2.98%	db.r5.xlarge	98.55%	\$ 1,825.73	db.r5.xlarge	\$ 1,825.73
131	Development	db-mssql-shared-usw2-021-dev	sqlserver-ee 13.00.6300.2.v1	2.26%	db.r5.xlarge	21.29%	\$ 1,825.73	db.r5.xlarge	\$ 1,825.73
132	Development	db-mssql-shared-usw2-031-dev	sqlserver-ee 15.00.4043.16.v1	2.34%	db.r5.xlarge	14.46%	\$ 1,825.73	db.r5.xlarge	\$ 1,825.73
133	Development	tf-viacbs-biztalk-mule-dev	sqlserver-ee 12.00.6329.1.v1	1.60%	db.r5.xlarge	19.10%	\$ 1,825.73	db.r5.xlarge	\$ 1,825.73
199	Development	vc-ccs-tf-test-all-features-dev-ter	sqlserver-ee 15.00.4043.16.v1	1.24%	db.r5.2xlarge	13.17%	\$ 3,651.46	db.r5.xlarge	\$ 1,825.73
204	Development	vc-ccs-tf-test-all-features-dev-ter	sqlserver-ee 15.00.4043.16.v1	1.24%	db.r5.2xlarge	18.66%	\$ 3,651.46	db.r5.xlarge	\$ 1,825.73
206	Production	db-mssql-backup-test-prod	sqlserver-ee 15.00.4043.16.v1	5.28%	db.r5.xlarge	23.59%	\$ 1,825.73	db.r5d.xlarge	\$ 1,871.72
207	Production	db-mssql-biztalk-mule-use1-prod	sqlserver-ee 15.00.4043.16.v1	4.27%	db.r5.xlarge	31.02%	\$ 1,825.73	db.r5d.xlarge	\$ 1,871.72
220	Production	db-mssql-cognos-use1-032-prod	sqlserver-ee 13.00.6300.2.v1	2.31%	db.r5.2xlarge	23.05%	\$ 3,651.46	db.r5d.xlarge	\$ 1,871.72
221	Production	db-mssql-erpm-use1-prod	sqlserver-ee 15.00.4043.16.v1	2.35%	db.m5.xlarge	17.70%	\$ 1,705.28	db.r5d.xlarge	\$ 1,871.72

Right-sizing

3. Optimize RDS instances, storage, and Auto Scaling for cost

Compute optimization

Factors affecting RDS Compute Cost

Factors	Options
Pricing/billing model	On-demand DB instances vs Reserved DB instances
RDS instance utilization	Under-utilized vs. over-utilized
High-Availability setup	Single-AZ vs Multi-AZ
Instance runtime	Stop or start DB instances for predictable workload
RDS instance type	Memory optimized (r, x, or z class) vs. standard (m class) vs. burstable (t class)

Optimize the instance type and cost

Instance class	Instance family	Memory to CPU	vCPU	RAM	Network (Gbps)	EBS optimized (IOPS/BW Gbps)	Instance storage (TB)	2xlarge SQL Server Standard Edition*	Notes
Burstable	t3	4 to 1	8	32 GB	Up to 5	15,700/2.8	---	\$2.088	Low cost, non-prod, or lower environments
General purpose	db.m5	4 to 1	96	384 GB	25	80,000/19	---	\$2.548	Supports RDS Custom
	db.m5d	4 to 1	96	384 GB	25	80,000/19	3.6	\$2.581	TempDB on NVMe
	db.m6i	4 to 1	128	512 GB	50	160,000/40	---	\$2.548	Supports RDS Custom
Memory optimized	db.z1d	8 to 1	48	384 GB	25	80,000/19	---	\$3.419	4 GHz all-core
	db.r5	8 to 1	96	768 GB	25	80,000/19	---	\$3.040	Supports RDS Custom
	db.r5d	8 to 1	96	768 GB	25	80,000/19	3.6	\$3.142	TempDB on NVMe
	db.r5b	8 to 1	96	768 GB	25	260,000/60	---	\$3.175	I/O-bound workload
	db.r6i	8 to 1	128	1 TB	50	160,000/40	---	\$3.040	Supports RDS Custom
	db.x1e	30 to 1	128	4 TB	25	80,000/14	---	\$4.943	Memory intensive workloads
	db.x2iedn	32 to 1	128	4 TB	100	80,000/80	3.8	\$5.263	TempDB on NVMe

Instance Scheduler on AWS (RDS Only)

AWS-provided solution

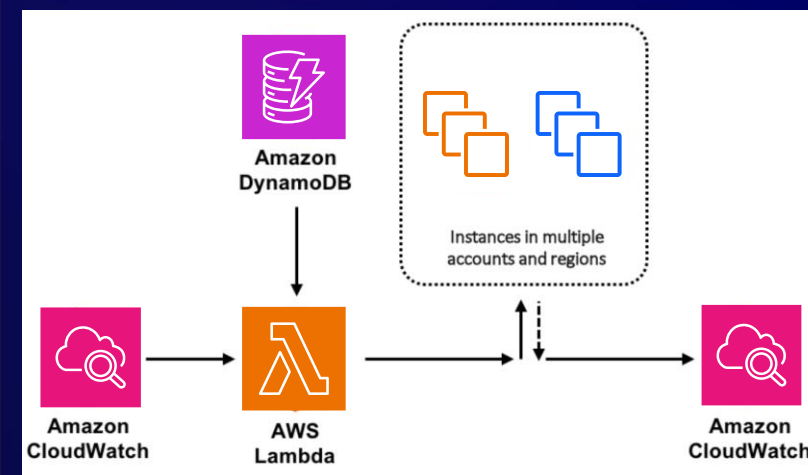
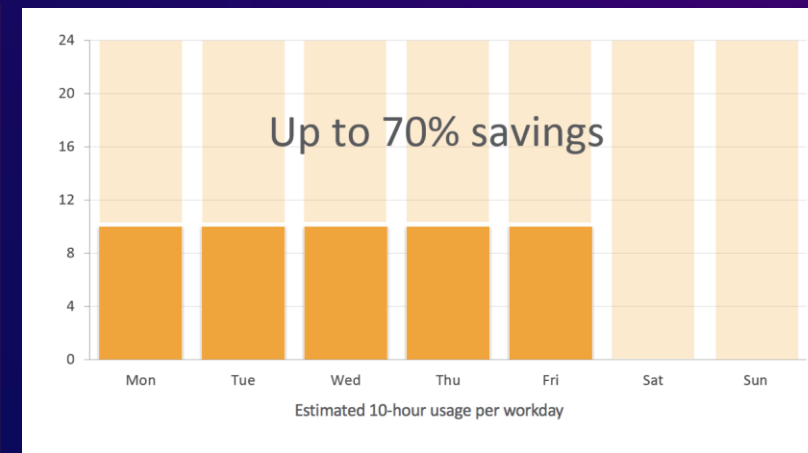
- Custom start and stop schedules
- Works with EC2 and RDS instances
- Deploy using AWS CloudFormation

Selectively tag instances to schedule

- Multiple schedules per instance
- 5-minute granularity

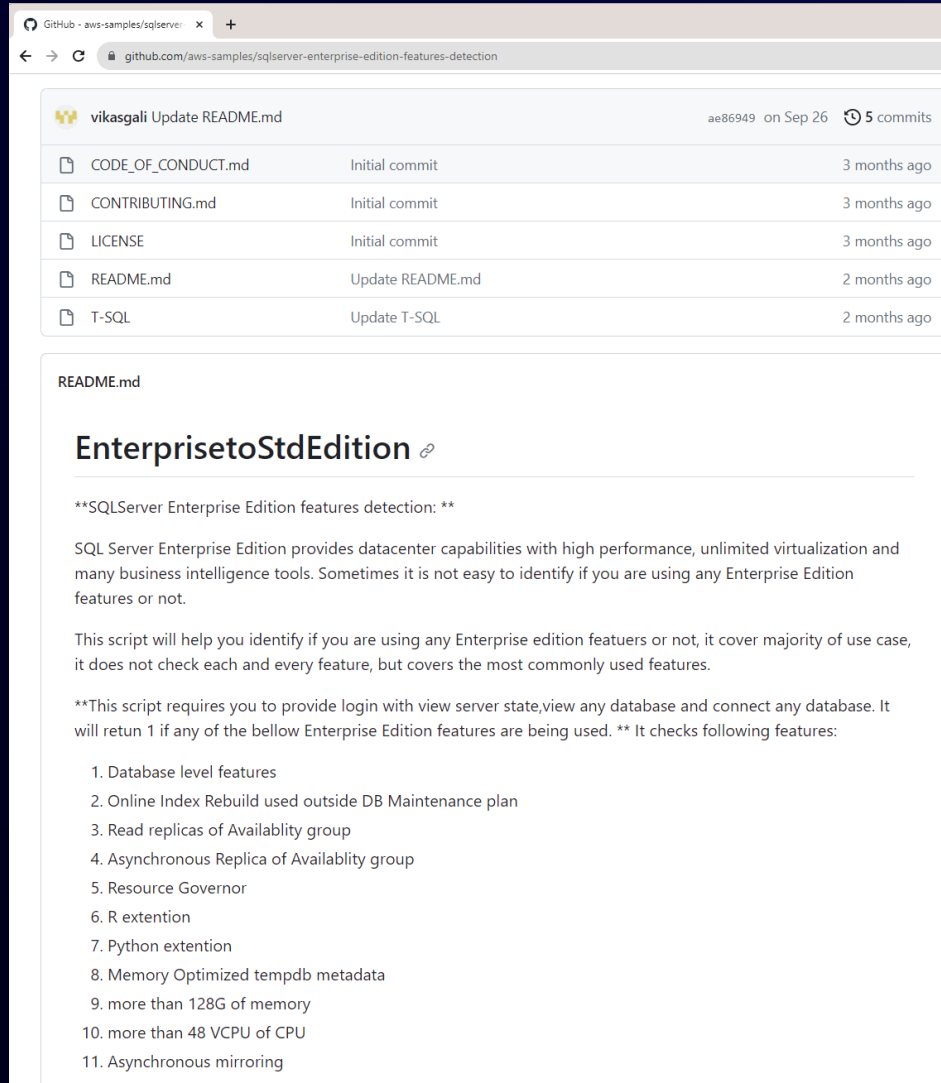
Use AWS Lambda to resize instances (RDS and RDS Custom)

- Deploy Lambda functions for scaleup and scaledown using a schedule



<https://aws.amazon.com/answers/infrastructure-management/instance-scheduler/>

SQL Server edition optimization



GitHub - aws-samples/sqlserver: vikasgali Update README.md ae86949 on Sep 26 5 commits

File	Commit	Time
CODE_OF_CONDUCT.md	Initial commit	3 months ago
CONTRIBUTING.md	Initial commit	3 months ago
LICENSE	Initial commit	3 months ago
README.md	Update README.md	2 months ago
T-SQL	Update T-SQL	2 months ago

README.md

EnterprisetoStdEdition [🔗](#)

****SQLServer Enterprise Edition features detection: ****

SQL Server Enterprise Edition provides datacenter capabilities with high performance, unlimited virtualization and many business intelligence tools. Sometimes it is not easy to identify if you are using any Enterprise Edition features or not.

This script will help you identify if you are using any Enterprise edition featurers or not, it cover majority of use case, it does not check each and every feature, but covers the most commonly used features.

****This script requires you to provide login with view server state,view any database and connect any database. It will retun 1 if any of the bellow Enterprise Edition features are being used. **** It checks following features:

1. Database level features
2. Online Index Rebuild used outside DB Maintenance plan
3. Read replicas of Availability group
4. Asynchronous Replica of Availability group
5. Resource Governor
6. R extention
7. Python extention
8. Memory Optimized tempdb metadata
9. more than 128G of memory
10. more than 48 VCPU of CPU
11. Asynchronous mirroring

This script will help you identify if you are using any Enterprise edition features or not

<https://github.com/aws-samples/sqlserver-enterprise-edition-features-detection/>



4. Discussion

Thank you!

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Please complete the session
survey in the mobile app

